

Zeyu Lai

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Education

University of California, San Diego 2025 – 2029
Undergraduate in Data Science & Electrical Engineering; Minor in Mathematics

BASIS International School Park Lane Harbour 2022 – 2025
High School Diploma | GPA: 3.79/4.0

Experiences

CoCo Matrix July 2026 – September 2026
Robotics Infrastructure Engineer Intern | Shanghai, China

- Designed unified multi-robot infrastructure for GR3, dual-PiPER-X, and Atribot S1, integrating policy serving, perception, teleoperation, safety, and hardware execution through modular Dora dataflow graphs.
- Implemented latency-aware asynchronous VLA inference and RTC action-chunk scheduling, overlapping inference with execution while maintaining temporal alignment, smooth chunk transitions, and control-rate stability.
- Led the WRC ICL runtime and on-robot deployment, alongside an agentic VLM harness for multi-stage task orchestration, visual outcome verification, and closed-loop recovery.
- Developed trajectory-processing and DAgger/HITL systems, including jerk-limited motion smoothing, cross-robot IK retargeting, MODEL/EXPERT/HOLD arbitration, and synchronized intervention-data collection.
- Established contract-driven model onboarding and deployment workflows with automated qualification, operator interfaces, runtime watchdogs, and fail-closed hardware execution.

Hao Su's Lab Mar 2026 – Present
Undergraduate Research Assistant (Embodied AI research group) | UC San Diego
Developing dexterous teleoperation frameworks, sim-to-real pipelines, and predictive world models.

MURO Lab Oct 2025 – May 2026
Undergraduate Research Assistant (Distributed Robotics and Multi-Agent Coordination) | UC San Diego
Researched distributed robotics and multi-agent coordination; Developing multi-robot map-merging infrastructure to enable autonomous navigation for robot fleet (TurtleBot4 platforms) and unified it into a global representation.

TRITON-AI Oct 2025 – May 2026
Path Planning Team Member (Autonomous Vehicles) | UC San Diego
Worked on LiDAR-camera sensor fusion for F1TENTH and contributed to the path planning team for Triton-AI's autonomous go-kart.

Publications

MSV-WAM: Multi-Scale Visual Streams for Long-Horizon World Action Modeling [coming]
Xingyu Wang, **Zeyu Lai**, Jiayu Dong, Jiaxin Fang, Weikang Wan, Yuxiang Gao, Mengdi Xu
Under Review, ICLR 2027

- BridgeWAM: Learning a Latent Bridge between World Modeling and Action Generation** [coming]
 Jiaxin Fang, Ruitong Liu, Xingyu Wang, **Zeyu Lai**, Weikang Wan, Mengdi Xu, Yuxiang Gao
Under Review, ICLR 2027
- DexSeed: Scalable Physics-Grounded Demonstration Generation for Sim-to-Real Dexterous Manipulation** [coming]
 Weikang Wan, Jinqiu Wang, Jianyu Hou, **Zeyu Lai**, Henrik I Christensen, Hao Su
Under Review, ICLR 2027
- DexFLEX: Contact-Aware Foundation Controller for Command-Guided Dexterity** [Website]
 Weikang Wan, **Zeyu Lai**, Jianyu Hou, Yuchen Zhou, Jiawei Fu, Henrik I Christensen, Hao Su
Conference on Robot Learning (CoRL) 2026
- Diving into the virtual realm: Exploring the mechanics of virtual reality** [ResearchGate]
Zeyu Lai
Proc of the 2023 International Conference on Machine Learning and Automation

Projects

- s-motf: Latent World-Action Model (L-WAM)** 2026
 Architected a 50 Hz proprioceptive quadruped controller (Unitree Go1, MuJoCo) around a Mixture-of-Transformers backbone with a 3-step rectified flow-matching action head and prior/posterior latent-plan alignment. Distilled three PPO specialists — locomotion, hind-leg balance, fall recovery — into a single command-conditioned policy retaining 83–96% of specialist return, where an equal-capacity MLP baseline collapsed on the multimodal skill ($4.7\times$ lower return, $4.1\times$ shorter survival).
- Baton: Multi-Agent Handoff & Orchestration Framework** 2026
 Engineered a local control framework to hot-swap active CLI coding sessions (Claude Code $\leftrightarrow$ Codex) mid-task. Captured terminal streams via node-pty/xterm.js and distilled unverified workspace states into provider-neutral handoff packets to bypass rate limits without task re-explanation.
- DPPO for Dexterous Manipulation** 2026
 Implemented Diffusion Policy Policy Optimization (DPPO) for high-dimensional, state-based dexterous manipulation tasks (Adroit Hand). Developed a modular pipeline featuring human demonstration pre-training via behavior cloning, action-chunking, and fixed expert state normalization, bridging generative diffusion models with RL by fine-tuning a pre-trained diffusion policy to maximize task rewards using PPO.
- RL Gymnasium** 2026
 Developed a clean, extensible reinforcement learning repository for training and evaluating classic Gymnasium control tasks, utilizing Python, PyTorch, and reinforcement learning principles for algorithmic implementation.
- Uniq: Intelligent Browser Automation System** 2026
 Engineered an autonomous browser agent using Node.js, Playwright, and Gemini 3.0 Flash for task planning and decision making. Developed a resilient Agent Loop featuring semantic auto-scroll, graduated failure recovery, and DOM-based interactive region detection using TypeScript and WebSockets.

Technical Skills

Languages: Python, C, Java, JavaScript
Libraries & Tools: NumPy, CUDA, OpenCV, MediaPipe, ROS2, micro-ROS, SLAM Toolbox, PyTorch, Scikit-Learn, Isaac Gym, Isaac Lab, Isaac Sim, Gymnasium

Hardware: GR3, dual PiPER-X, Astribot S1, X-ARM7, LEAP HAND, ESP32, Arduino, Turtlebot4, Jetson Nano, Sensor integration (LiDAR, IMU, etc.)

Machine Learning & AI: Deep learning architectures, Reinforcement learning, Imitation Learning, VLA, WAM, Multi-Robot Policy Infra

Design & Dev Tools: PCB Design, Fusion 360, Jupyter Notebook, Google Colab, Conda/uv, K8S, GitHub Actions, Linux

Honors & Certifications

Honors:

- 2024 USACO Gold Division
- 2024 ICC Finalist
- BASIS China Hackathon 1st Place (1/36)
- Euclid Math Contest Distinction (top 20%)
- IMMC 2024 Greater China Region Finalist

Certifications (Stanford University):

- Supervised ML: Regression & Classification
- Advanced Learning Algorithms
- Unsupervised Learning, Recommenders, RL